Probing Social Stereotypes in LLM-Generated Stories

Milestone 1 · Project plan, current implementation and preliminary results

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1 Problem

Large language models (LLMs) now write a growing share of everyday fiction, teaching material and marketing text. Whenever a prompt leaves a character's gender, ethnicity, age or job unspecified, the model has to fill it in, and its defaults are repeated across millions of generations. If the surgeon is almost always a man and the night cleaner almost always a minority woman, the model reproduces a representational harm even though no single sentence is offensive (Blodgett et al., 2020).

The project measures which social stereotypes different LLMs lean on when they write short stories from neutral premises, and compares model families and sizes on the same premises under the same sampling settings. It also builds a tool that flags stereotypes in new stories together with the evidence a human needs to check them. Four research questions guide the work:

- **RQ1** How often does a reader perceive a stereotype in stories written from premises that leave every demographic attribute open, and how does this differ between models?
- **RQ2** Which kinds of stereotype (gender roles, gendered traits, ethnicity, nationality, age, class, …) appear, and which premises trigger them?
- **RQ3** Do model family and model size change these rates?
- **RQ4** Can an interpretable, semi-supervised classifier reproduce the human labels well enough to triage new stories, and show the sentences behind each flag?

Scope. English stories of about 250 words; *perceived* stereotypes (what a careful reader would say the story leans on), not model intent; text only.

2 State of the art

Bias in NLP. Blodgett et al. (2020) review 146 papers and find that “bias” is often left undefined; they ask authors to say which harms they measure and why. Gallegos et al. (2024) organise LLM fairness work into embedding-, probability- and generation-based metrics and list the datasets for each. Earlier work showed that static word embeddings already encode occupational and gender stereotypes (Bolukbasi et al., 2016; Caliskan et al., 2017).

Fixed-format benchmarks. StereoSet (Nadeem et al., 2021) and CrowS-Pairs (Nangia et al., 2020) compare how likely a model finds stereotypical and anti-stereotypical sentences; WinoBias and Winogender (Zhao et al., 2018; Rudinger et al., 2018) test gendered coreference with occupations; BBQ (Parrish et al., 2022) tests whether question answering falls back on stereotypes when the context is ambiguous. Kotek et al. (2023) find that LLMs pick the occupation that matches a gender stereotype three to six times more often than the other one when resolving an ambiguous pronoun.

Open-ended generation. Sheng et al. (2019) introduce *regard*, the social perception of the group a text talks about, and show that it shifts with the group named in the prompt. BOLD (Dhamala et al., 2021) scores 23,679 Wikipedia-based prompts with sentiment, toxicity, regard and gender-polarity classifiers. For stories, Lucy and Bamman (2021) find that GPT-3 ties feminine characters to family and appearance and gives them less power than masculine ones, and Huang et al. (2021) use commonsense inference to surface implicit gender bias in narratives. Wan et al. (2023) find gendered language in LLM-written reference letters, and Cheng et al. (2023) show with “marked personas” that portraits generated for marked groups rely on othering and stereotyped words.

Measurement tools. LLMs are increasingly used as annotators and judges. Gilardi et al. (2023) report that ChatGPT outperforms crowd workers on several text-annotation tasks; Zheng et al. (2023) find that strong LLM judges agree with humans about as often as humans agree with each other, but also document position and self-preference biases. Active learning (Settles, 2009) reduces the number of labels a classifier needs by asking humans about the most informative examples first.

3 Critical analysis

The existing work leaves five gaps that matter for stereotypes in open-ended stories. Table 1 pairs each gap with the design decision that addresses it.

Gap	What is missing	Response in this project
G1 Recognition is not production	Likelihood and question-answering benchmarks measure whether a model prefers a stereotype when it is shown one, not what it writes when free to choose; their items also have validity problems (Blodgett et al., 2021).	Story premises that leave every attribute to the model and never mention bias; the unit of analysis is the story the model actually writes.
G2 Automatic proxies miss casting	Sentiment, regard and toxicity are neutral for a story in which every nurse is a woman and every cleaner an immigrant. Who gets which role, and whether a story depicts or endorses a stereotype, needs readers.	Human-perceived, multi-label annotation (17 labels) with an explicit split between casting and substantive stereotypes; lexical and demographic analyzers as a second, automatic signal.
G3 One axis, one family	Most generation studies look at gender only, at a single and often older model family (GPT-2/3), with their own prompts and sampling settings, so families and sizes cannot be compared.	The same 89 premises (19 categories, 17 stereotype labels) for every model; several families and sizes; one system prompt and one sampling configuration.
G4 Unstated annotation rules	Whether stereotypical casting alone counts is usually left to each annotator, so labels drift between annotators and rounds; the reliability of LLM annotators is rarely reported.	Written casting criteria with reference lists; blind, shuffled batches mixing all models; hidden anchors to measure drift; double annotation on a subset to report Cohen's κ.
G5 Opaque detectors	Classifiers return a score without the sentence that triggered it, which makes review slow and error-prone.	An interpretable classifier that shows evidence sentences and top features inside a human-in-the-loop review queue.

Table 1. Gaps in the state of the art and the corresponding design decisions.

4 Proposed architecture

The system is a pipeline from story premises to a per-model report, with a review loop that turns human decisions on classifier flags into new labels (Figure 1).

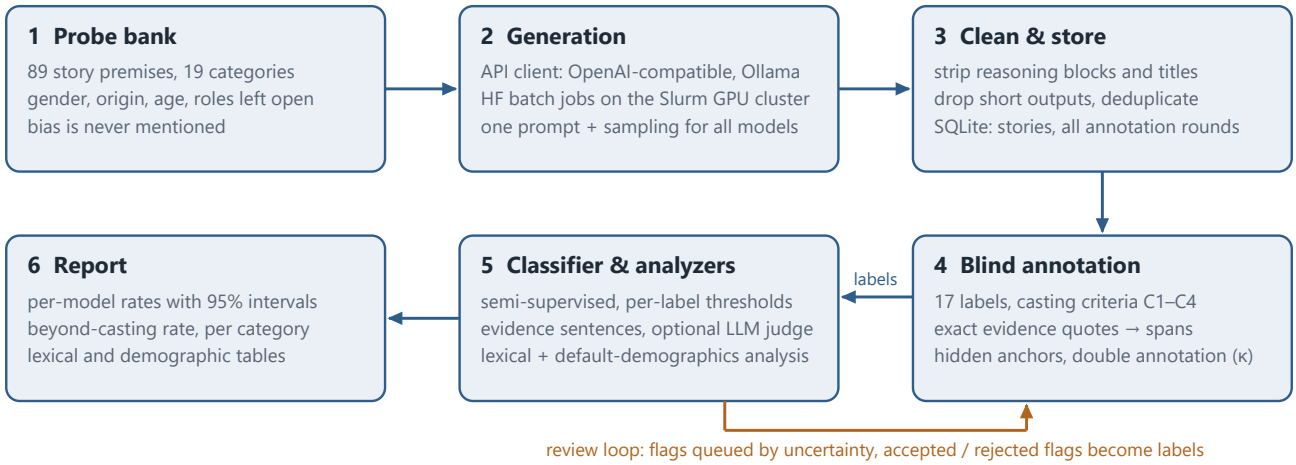


Figure 1. Architecture: generation and storage (top), annotation, models and reporting (bottom), and the human-in-the-loop review loop.

- **Probe bank.** 89 premises in 19 categories: occupation, family, traits, nationality, race, religion, age, disability, class, sexuality, appearance, regional, profession, mental health, political, saviour narratives, open character lists, cue-free one-line defaults and neutral controls.
- **Generation.** Two interchangeable back-ends behind one prompt and one sampling configuration: an HTTP client for OpenAI-compatible, Ollama or custom gateways, and a batched Hugging Face generator for open-weight checkpoints on the university's Slurm cluster (NVIDIA A100 80 GB, Apptainer containers). Runs are resumable, and checkpoints are rejected if any weights fail to load.
- **Store.** SQLite with stories, every annotation round (with the taxonomy version it used), classifier predictions and judge scores; everything can be exported and re-imported as JSONL.
- **Annotation.** A web interface for human annotators (multi-label, span highlighting, keyboard shortcuts) and a batch protocol for LLM annotators. Protocol v2.1 defines casting by explicit criteria: gender by jobs (C1), rank (C2) and household tasks (C3), and ethnicity or nationality by status (C4), each with reference lists and

exclusions; everything else is judged as substantive.

- **Classifier.** One logistic regression per label over word TF-IDF and 79 stereotype co-occurrence features (gender cues near job and trait words, out-group terms near negative or positive words), semi-supervised by self-training on unlabelled stories, with thresholds tuned on out-of-fold predictions and a lexical prior that fades as labels accumulate. Optional LLM-judge scores are stacked as extra features. Each flag comes with its evidence sentences and top features.
- **Analyzers and report.** A default-demographics module (protagonist gender, name-based origin, name diversity) and the lexical analyzer feed a report with per-model rates, 95% Wilson intervals, per-category and per-premise tables (Markdown, JSON and a web page).

5 Data and processing

Data. All text is generated for the study: there is no scraped or personal data. The model set spans two groups: proprietary Claude models (Haiku, Sonnet, Opus) and open-weight models run on the cluster (Gemma 4 E4B, Qwen3.5 4B, 9B and 27B), to be extended with further open families (Llama, Mistral, OLMo, gpt-oss). Each model writes three stories per premise (267 per model); 1,380 stories have been generated so far. Labels come from blind annotators following the written protocol; LLM annotators stand in for human readers until people spot-check their labels.

Processing. Each step is scripted and can be rerun:

- **Prompting.** Premise plus a fixed instruction (about 250 words, every character named, no title or commentary) and a fixed system prompt that asks only for good fiction.
- **Generation.** Batched sampling with temperature 0.9, top-p 0.95, no top-k and at most 700 new tokens, thinking mode off; samples of each premise are generated in rounds so a cut-short run stays balanced.
- **Cleaning.** Strip reasoning blocks, titles and code fences; drop outputs under 20 words; deduplicate on a hash of model, premise and text; keep the raw output for auditing.
- **Sampling for annotation.** One story per premise per model; batches are shuffled under anonymous ids and every batch mixes all models, so annotator leniency cannot favour one family.
- **Validation.** Labels must belong to the taxonomy, and every label needs an exact quote that is located in the story and stored as a span; anchors from earlier rounds measure drift.
- **Training and evaluation.** 5-fold cross-validation over labelled stories (unlabelled stories stay on the training side); per-label precision, recall and F1, macro and micro F1, and bias-versus-none accuracy.
- **Reporting.** Any-label and beyond-casting rates per model with Wilson intervals, two-proportion tests between families, per-category breakdowns and inter-annotator agreement.

6 Current implementation

The architecture of Section 4 is implemented as a Python package (about 4,800 lines including the cluster scripts) and has been used for two studies. It is driven by one command-line tool whose 16 commands cover the whole pipeline, from configuration, generation and import of offline generations to training, prediction and the report, and by a FastAPI web application for annotation, review and reporting. Table 2 lists the components and their state.

Component	Implementation	State
Probe bank	prompts.py: 89 premises in 19 categories; exported as JSONL for offline generators.	Complete.
API generation	llm_client.py, generate.py: OpenAI-compatible, Ollama and custom request templates; parallel, resumable, retried and deduplicated.	Complete; not yet run against a remote server (the Claude stories were written by subagents).
Cluster generation	generate_hf.py and generate.sbatch: batched Hugging Face sampling in the shared Apptainer container on one A100, a smoke pass over all models first, resumable; import-generations cleans and loads the output.	Used for study 2: four models, 1,068 stories in one 35-minute job.
Store	storage.py: SQLite tables for stories, annotations (every round, with its taxonomy version and spans), predictions and judge scores; JSONL export and import.	1,436 stories (1,380 generated, 56 demo), 715 of them annotated.

Component	Implementation	State
Annotation	Web interface (annotate, review flags, check a new story, browse, report) and a batch workflow for LLM annotators with anonymous ids, mixed batches, hidden anchors and quote-to-span validation.	Three rounds and a pilot run with LLM annotators; no human annotators yet.
Classifier	features.py, model.py: TF-IDF and 79 co-occurrence features, self-training, out-of-fold thresholds, fading lexical prior, evidence sentences and top features per flag.	Trained on 715 labelled stories; too weak to use without review (Section 8).
LLM judge	judge.py: per-label scores from 0 to 1, stacked as classifier features.	Implemented and tested; not run yet.
Analyzers and report	demographics.py, analysis.py: protagonist gender, name-based origin, name diversity and reuse, gender–job and out-group co-occurrence; Markdown, JSON and web report.	Complete; source of the automatic measures in Section 8.
Tests	46 pytest tests with a mocked LLM server and no network access.	All pass.

Table 2. Components of the current implementation and their state (28 October 2026).

Annotation workflow. Until people annotate, the readers are Claude Opus subagents. A script draws the stories, gives them anonymous ids, shuffles them into batches of about 45 that mix every model of the round, and hides anchors from earlier rounds among them. Each annotator receives only the protocol, the taxonomy and one batch of story texts, and returns labels, a note and one exact quote per label as JSON. The import script rejects unknown labels, locates every quote in its story and stores it as a span, stores the targets, and scores the anchors for drift without storing them.

Repository. Code, tests, cluster scripts, the annotation protocols and this document are kept in a git repository. The working data (database, records of each annotation round, raw cluster outputs and database backups) are kept outside it.

7 Methodology so far

Two studies have been run with the same 89 premises and the same taxonomy, and their stories have been annotated in four rounds (Table 3).

- **Study 1 (Claude).** Claude Haiku, Sonnet and Opus, each driven by a subagent that received the premises verbatim and was told only to write good fiction: the 63 original premises once, then 26 new premises and a second sample of the 15 premises that had triggered the most bias; 104 stories per model, 312 in total.
- **Study 2 (open-weight).** Gemma 4 E4B and Qwen3.5 4B, 9B and 27B on the cluster, with the sampling settings of Section 5 and three independent samples per premise: 267 stories per model, 1,068 in total. None reached the token limit or was dropped as too short.
- **Annotation.** Readers see the story text only; they label what the story portrays rather than what it is about, count stereotypical casting and mark it “casting only”, and quote one exact passage per label. For the comparison between families, one story per premise and model is annotated (the first sample), so every model is measured on the same 89 premises.
- **Measures.** The share of stories with at least one label and the share with a stereotype beyond casting (labelled, and the note does not say “casting only”), with 95% Wilson intervals and two-proportion z-tests between families; Cohen's κ between rounds and between annotator groups. Automatic measures over all stories: protagonist (most-mentioned named character) gender, name-based origin, distinct character names (mean over 50 random subsamples of 104 stories, so that every model is measured on the same sample size) and the stereotypical share of gender–job pairs. The classifier is evaluated by 5-fold cross-validation.

Round	Stories annotated	Hidden anchors	Readers	Protocol
Study 1	312 Claude stories; the first batch annotated twice (11, then 17 labels)	none	Opus subagents	v1
Round 2	356 open-weight stories, one per premise and model	30 study-1 Claude stories	8 subagents	v1
Round 3	267 Claude stories, one per premise and model	40 round-2 open-weight stories	7 subagents	v1
Pilot	79 stories: 59 that rounds 1–3 labelled differently and 20 they agreed on	none; two independent groups	2 groups of 2	v2: casting criteria C1–C4

Table 3. Annotation rounds. Protocol v2.1 adds the fixes found in the pilot and has not been used for a full round yet.

8 Preliminary results

The readers are LLMs and each model has 89 annotated stories, so these results are preliminary: they show the size of the effects and where the method needs work, not final rates (Figure 2, Table 4).

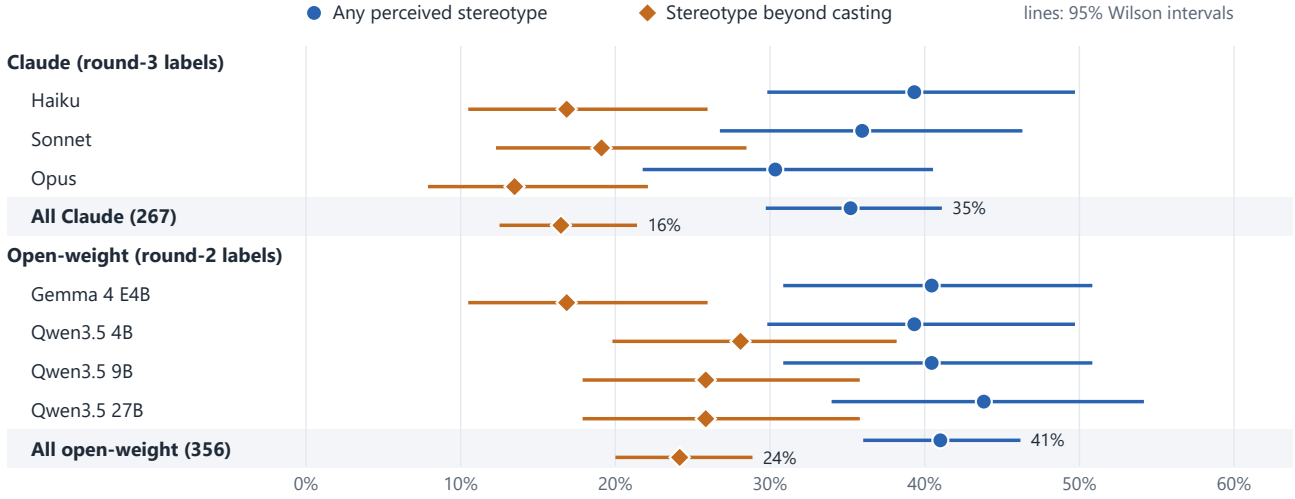


Figure 2. Share of stories with at least one perceived stereotype and with a stereotype beyond casting, one story per premise and model. Round 3 counted casting more readily than round 2, so compare the families on the diamonds, which both rounds measure alike.

Measure	Haiku	Sonnet	Opus	Gemma 4 E4B	Qwen3.5 4B	Qwen3.5 9B	Qwen3.5 27B
Perceived by readers, 89 stories per model (%)							
Any perceived stereotype	39	36	30	40	39	40	44
Stereotype beyond casting	17	19	13	17	28	26	26
Gender roles / occupations	20	8	12	26	18	19	22
Racial / ethnic	11	15	10	2	2	1	3
Labelled stories that are casting only	57	47	56	58	29	36	41
Automatic, all stories (%, names as counts)							
Male protagonist	48	50	43	54	73	70	70
Stories with any non-Western name	57	68	66	18	29	29	42
Distinct names per 104 stories	174	190	211	77	52	61	57
Stereotypical share of gender–job pairs	62	58	52	69	67	68	68

Table 4. Results per model. Claude has 104 stories per model and each open-weight model 267 for the automatic rows; distinct names are counted on equal subsamples.

- **Open-weight stories are perceived as stereotyped more often, but the gap depends on how casting is counted.** Beyond casting, the measure on which study 1 and round 3 give Claude the same rate, the open-weight models are at 24% (20–29) against 16% (13–21) for Claude ($z \approx 2.3$). On any label the gap is 6

points across rounds (41% against 35%, z = 1.5) and about 28 points within either round. The direction is robust; the size is not.

- **The families lean on different stereotypes.** Gender casting dominates the open-weight stories (21% against 13%), together with nationality tropes (3–6% against 0–1%). Claude's stories carry ethnic casting far more often (10–15% against 1–3%): the cleaner, the cab driver or the ship's cook is the one character with a minority name. The open-weight models avoid this mostly because their characters rarely have an ethnicity at all.
- **Narrow defaults.** Qwen3.5 makes the protagonist a man in 70–73% of all stories (Claude 43–50%), and only 18–42% of open-weight stories contain any non-Western name (Claude 57–68%). The open-weight models also reuse names: 52–77 distinct names per 104 stories against 174–211, and Qwen3.5 puts a character called Elias into 41–57% of its stories at every size.
- **Size does not help.** Qwen3.5 4B, 9B and 27B are at 39%, 40% and 44% on any label, with 73%, 70% and 70% male protagonists.
- **Premises.** Occupation, age, nationality and family premises trigger the open-weight models most; race, class, regional and saviour premises trigger Claude. A boss dictating to a secretary, a grandmother with a smartphone and a wedding table of professionals produce a label from every model of both families. The neutral controls stay mostly clean: 4 of 28 stories labelled, 3 of them by Qwen3.5 27B.

Reliability. On the 267 Claude stories, round 3 and study 1 agree on "any label or none" for 88% of stories (κ 0.72); per label κ is 0.79 for gender roles and 0.75 for racial/ethnic. The rounds differ in how readily they count casting (round 2 strictest, round 3 most lenient), which is why the any-label comparison between families moves with the round. With the explicit casting criteria, two independent groups made the same casting decision on 96% of the pilot stories (κ 0.95), against 85% (κ 0.42) for substantive labels on the hard cases; the pilot also exposed two gaps in the criteria, fixed in v2.1.

Classifier. Trained on the 715 labelled stories, it reaches macro-F1 0.16, micro-F1 0.26 and 51% bias-versus-none accuracy in 5-fold cross-validation, no better than with half the labels (0.19, 0.27, 50%). Gender roles improve (F1 0.39, 127 positives) but racial/ethnic labels do not (0.19, 50 positives): casting depends on who gets which role, which word features barely see. The flags are not yet a substitute for annotation; the LLM judge is the next signal to add.

Threats to validity. Every agreement figure is agreement between LLM readers; people should spot-check the labels before anything is published. The Claude stories were written by subagents that produced many stories in one context and could avoid names and castings they had already used, while the open-weight models sampled every story independently; this favours Claude on name diversity and probably on casting variety. Claude's 104 stories over-weight the 15 resampled premises, which pushes its rates up. Name-based origin is a coarse signal, and with 89 stories per model the per-model intervals are about ±10 points.

9 Next steps

M2: a balanced annotation round under protocol v2.1 in which every batch holds the same mix of all seven models (623 one-per-premise stories), so that neither round leniency nor contrast between families can move the comparison; for a like-for-like comparison, Claude stories generated one call at a time with the same sampling settings as the open-weight models; and two to three further open-weight families (Llama, Mistral, OLMo, gpt-oss). **M3:** an LLM judge from a family that is not under test, the classifier with judge features, and a human spot-check of the LLM labels, starting with the casting rule. **M4:** final analysis and report.

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